

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION

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Project Work Phase-1 (EC7PW1) Synopsis

on

“PNEUMONIA DETECTION USING MACHINE LEARNING”

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SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY

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Project Work Phase-1 (EC7PW1)

Synopsis on “PNEUMONIA DETECTION USING MACHINE LEARNING”

Pneumonia is a leading cause of morbidity and mortality worldwide, particularly among children and the elderly. The number of individuals suffering from pneumonia is approximately more than 450 million a year. It is 7% of the overall population around the globe. Each year more than four million people die from Pneumonia. Pneumonia disease is prevalent among young children below 5 years old. Early detection and prompt treatment of pneumonia are crucial for preventing complications and improving patient outcomes. Chest X-ray imaging is commonly used for the diagnosis of pneumonia due to its accessibility and effectiveness in detecting abnormalities in lung tissue. However, the interpretation of chest X-ray images can be challenging and time-consuming, requiring expertise from radiologists. Automated methods based on deep learning techniques offer a promising solution to streamline the diagnosis process and improve the efficiency and accuracy of pneumonia detection.

Pneumonia is a severe respiratory infection that can cause life-threatening complications if left untreated. Although an early and accurate diagnosis is essential for successful treatment, conventional techniques of diagnosis can be expensive and time-consuming. For the automated detection of pneumonia from medical images, deep learning algorithms and computer vision methods have recently been investigated. The system automatically identifying pneumonia from chest X-ray images using deep learning algorithms. In order to categorize chest X-ray pictures as either pneumonia-positive or pneumonia-negative, creation of a convolutional neural network (CNN) model that will be trained on a dataset of chest X-ray images .

Developing a machine learning- based system for the early and accurate detection of pneumonia from chest X-ray images. Pneumonia, a prevalent respiratory infection, requires timely diagnosis for effective medical intervention. The study utilizes a diverse dataset containing both normal and pneumonia affected cases, ensuring comprehensive model training and evaluation. The methodology involves extracting relevant image features from chest X-rays, encompassing texture, intensity, and shape descriptors and showcase the

potential of machine learning models in pneumonia detection, with a focus on practical applicability in clinical settings. It provides the healthcare professionals a reliable tool that aids in the early identification of pneumonia cases, contributing to improved patient outcomes. The system has the potential to greatly improve the effectiveness and accuracy of pneumonia diagnosis and it is a useful instrument for the medical community because of its capacity to provide early and accurate diagnosis, which may be able to save lives and enhance patient outcomes to a considerable degree.

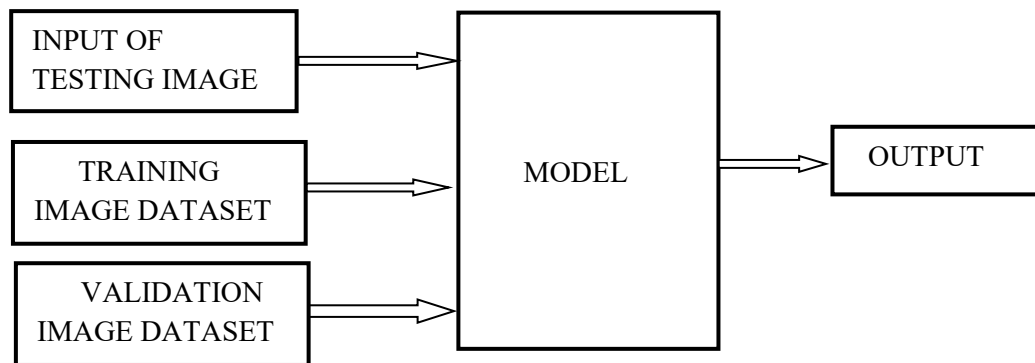


Figure 1: Proposed Model

The system involves three steps as in Figure 1. The issues of the design and implementation of the PNEUMONIA DETECTION USING MACHINE LEARNING can be subdivided into two main parts.

- ✓ The first part is Dataset: The database is further divided into three folders that is training data set, testing dataset, and validation dataset for each image category into Pneumonia / General.
- ✓ Data preprocessing: A series of data pre-processing methods were used to denoise and expand the chest X-ray images, thus enhancing the training of the object detection framework.
- ✓ Feature extraction: Feature extraction techniques are used to obtain a feature vector which is a reduced representation of the input data which contains information about the most important features of data. This feature vector is sent to a Convolutional Neural Network for classification of the output.

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